



Nutrient composition of *Mycetinis scorodonius* fruiting bodies and induction of sulfur-containing aroma compounds in surface mycelium

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Abstract

Mycetinis scorodonius is a garlic-scented mushroom from the family *Omphalotaceae* that can be used as a seasoning mushroom. γ -Glutamylmarasmin is considered the aroma precursor, which is converted into marasmin through enzymatic activity. Marasmin itself is unstable and breaks down into several other flavor compounds, such as methyl dithioformate, 2,3,5-trithiahexane and 2,4,5,7-tetrathiaoctane. This study examined the differences in aroma between fresh and dried fruiting bodies, as well as activation with water. It was found that fresh fruiting bodies activated with water had the highest aroma content, while dry fruiting bodies without the activation of water released almost no aroma compounds. The aroma content of surface mycelium at different ages (from 0 to 8, 8 to 16, and 16 to 22 days) was also examined, revealing a decrease in aroma compounds with increasing age. To identify potential precursors of γ -glutamylmarasmin, supplementation studies (L-cysteine, L-methionine, sulfate) were conducted with the mycelium, suggesting that L-cysteine could serve as a precursor. Although the mushroom is considered edible, its nutrient composition has not yet been studied. Since *M. scorodonius* was previously classified in the families *Tricholomataceae* and *Marasmiaceae*, the nutrient compositions determined here were compared with those of one mushroom from each of the three families. The greatest similarity was found with *Lentinula edodes* from the family *Omphalotaceae*, which corresponds to its current family classification.

Keywords Garlic-scented mushroom · *Omphalotaceae* · Mycelium age · L-cysteine · Marasmin

Introduction

Mushrooms have been an integral part of cuisine and medicine for centuries. They are not only low in calories but also rich in vitamins, antioxidants, glucans, proteins, and minerals (Manzi et al. 1999; Alam et al. 2009). However, these advantages are not the main focus when it comes to seasoning mushrooms. Seasoning mushrooms stand out because of their intense flavor and aroma. For example, there are mushrooms that taste like anise (*Agaricus arvensis*, *Clitocybe odora*, *Lentinellus cochleatus*); cloves (*Marasmius oreades*); chicken (*Laetiporus sulphureus*); or garlic (*Mycetinis*

scorodonius, *Mycetinis alliaceus*). Seasoning mushrooms are usually small and inconspicuous and are not suitable as a main ingredient. A very popular seasoning mushroom is the garlic-scented mushroom *Mycetinis scorodonius* (formerly known as *Marasmius scorodonius*). *M. scorodonius* belongs to the class *Agaricomycetes* and the order *Agaricales* (Breitenbach and Kränzlin 1991). In the pre-molecular era, when classification was still based on classical morphology, *M. scorodonius* belonged to the family *Tricholomataceae* (Wilson and Desjardin 2005). Since molecular technologies (internal transcribed spacer regions, ITS; large subunit, LSU) have been introduced, *Marasmius* has been classified in the family *Marasmiaceae* (Wilson and Desjardin 2005). However, ongoing phylogenetic research has led to its reclassification in the family *Omphalotaceae* and the genus *Mycetinis* (Wilson and Desjardin 2005; Oliveira et al. 2019). It is found in large parts of Europe, grows in summer and late summer until November, and has a strong aroma reminiscent of garlic (Breitenbach and Kränzlin 1991). It is said

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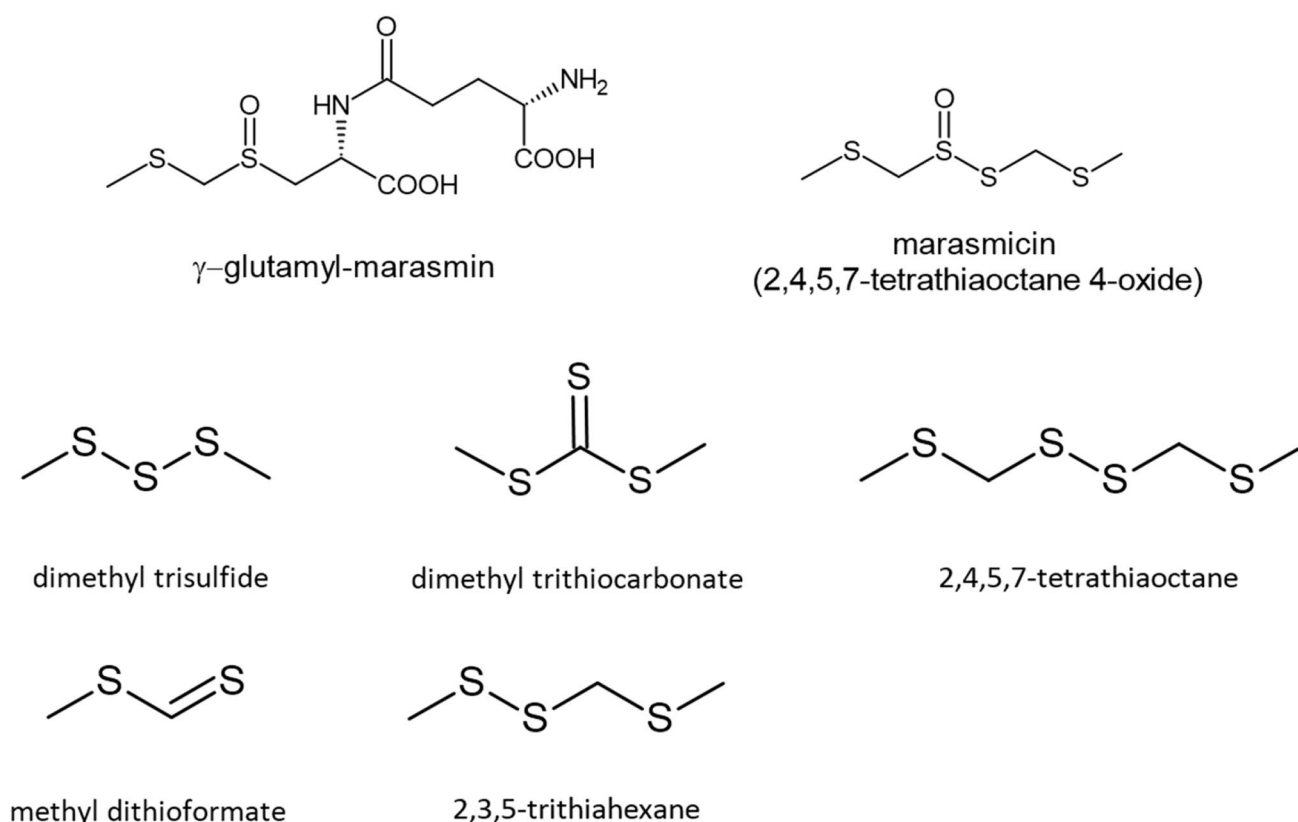


Fig. 1 Structures of the precursor γ -glutamylmarasmin, marasmicin, and the major flavor compounds found in *M. scorodonius* fruiting bodies

that the fruiting bodies are well suited for drying, as they do not lose their aroma during the drying process.

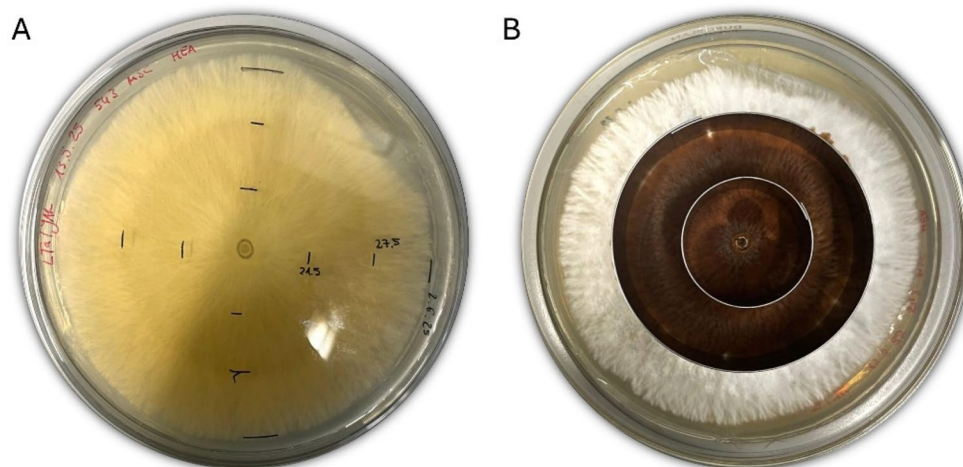
Unlike garlic (*Allium sativum*), where the characteristic odor is caused by allicin (diallyl thiosulfinate) and its breakdown products, in *M. scorodonius* the thiosulfinate marasmicin (2,4,5,7-tetrathiaoctane 4-oxide; Fig. 1) is responsible for the typical odor (Gmelin et al. 1976; Kusterer et al. 2011; Ahlborn et al. 2026). Marasmicin is formed by enzymatic degradation of the precursor γ -glutamylmarasmin (Fig. 1), which can account for up to 2.99% of dried *M. alliaceus* fruiting bodies, a close relative of *M. scorodonius* (Kusterer et al. 2011). Like garlic, the enzymes and substrates are located in separate cellular compartments, so the aroma compounds are released only after cell destruction. The aroma profile of *M. scorodonius* was recently analyzed by Ahlborn et al. (2026). Using headspace solid phase microextraction coupled with gas chromatography tandem mass spectrometry analyses (HS-SPME GC-MS/MS), we found that, in addition to dimethyl trisulfide, dimethyl trithiocarbonate, and 2,4,5,7-tetrathiaoctane, methyl dithioformate and 2,3,5-trithiahexane are the most important aroma compounds in dried and water-activated fruiting bodies. The flavor compounds were identified using RI, MS, olfactometry, and, when available, by comparison with a standard (Ahlborn et al. 2026). Methyl dithioformate was detected in

mushrooms for the first time. The aroma compounds found all had a garlic-like, onion-like, savory, sulfurous and/or cabbage-like smell and derive from marasmicin.

Observations from the late 1930s have shown that the intense smell of garlic does not apply to the mycelium, however (Badcock 1939). Badcock studied *M. alliaceus* and found that the originally white mycelium was odorless. After 3 weeks, the mycelium changed color from white to mars brown to seal brown around the inoculum. Compared to the white mycelium, the brown mycelium had an intense garlic odor. In older cultures, however, the white mycelium also had a slight garlic smell.

In the literature, no information is available on the aroma compounds of *M. scorodonius* mycelium of different ages, nor on the differences in aroma compounds between fresh and dried fruiting bodies, nor on the nutritional composition of its fruiting bodies; only very little is known about its relatives, such as *M. copelandii* or *M. alliaceus* (Badcock 1939; Oni et al. 2021). The aim of this study is therefore to describe the nutrient composition of *M. scorodonius* and to investigate differences in the aroma composition of fresh and dried fruiting bodies. In addition, the mycelium was examined at different ages and supplementation experiments were carried out to investigate the influence of possible precursors.

Fig. 2 Petri dish with mycelium of different ages. **A** Back of the Petri dish with marked growth zones. **B** Front of the Petri dish with stainless steel rings placed on it to divide the growth zones



Materials and methods

Cultivation of fungi and sample preparation

The *Mycetinis scorodonius* strain was purchased from Steintaler Edelpilz (Neu Wulmstorf, Germany) and cultivated on malt extract agar plates (2% malt extract and 1.5% agar-agar) for strain maintenance. For fruiting body production, liquid pre-cultures were prepared and used to inoculate the solid substrate, as described by Ahlborn et al. (2026). The fruiting bodies were freeze-dried using a laboratory freeze-dryer (Alpha 1–2 LD plus, Christ, Osterode am Harz, Germany; set to $-50\text{ }^{\circ}\text{C}$ and 0.34 mbar) ground into a fine powder and stored at $-20\text{ }^{\circ}\text{C}$ until further analysis.

In addition to the fruiting bodies, analyses were also carried out on the surface mycelium, with a focus on aroma analysis. A distinction was made between two main ideas: First, does the garlicky aroma of the mycelium depend on its age? And second, can the aroma of the mycelium be enhanced by adding sulfur-containing precursor molecules to the agar?

Preparing mycelium of different ages

To investigate the influence of age, large glass Petri dishes with an inner diameter of 13.5 cm were used. The surface of the agar was covered with a sterile piece of cellophane film cut to size (Serva Cellophane Sheets II (140 × 133 mm), Serva Electrophoresis GmbH, Heidelberg, Germany), on which a round piece (inner Ø, 3 mm) of agar from the stock plate was placed in the center. During growth on malt extract agar, radial growth zones (4, 8, and 12 cm) were marked on the back of the Petri dishes and growth was documented over time (Fig. 2A).

The mycelium of the fully overgrown agar plates was cut out using stainless steel rings of corresponding diameters (Fig. 2B), allowing the mycelium to be divided into three growth zones of different ages. The mycelium within these zones was carefully separated from the cellophane layer using a scalpel and then frozen at $-20\text{ }^{\circ}\text{C}$. The corresponding growth zones of five plates were combined. The mycelium was freeze-dried, stored at $-20\text{ }^{\circ}\text{C}$ until needed, and was then ground with a mortar and pestle prior to analysis.

Cultivation on supplemented media

To investigate whether the garlicky aroma can be enhanced by adding precursor molecules, malt extract media were supplemented with L-cysteine, L-methionine, and sulfate (as MgSO_4). The stock solutions for cysteine (0.75 M), methionine (0.167 M), and sulfate (1 M) were prepared with ultrapure water and sterile-filtered before being added to the autoclaved medium (final concentrations, 2% malt extract and 1.5% agar). Different final concentrations of supplements were examined: 0.5, 1, 2, 5, 15, and 50 mmol/L L-cysteine, 5 mmol/L L-methionine, and 5, 15, and 50 mmol/L sulfate. The supplemented medium was poured into Petri dishes with an inner diameter of 8.7 cm, covered with cut-to-size cellophane after solidification, and inoculated as described above. As a control, mycelium was cultivated on not supplemented malt extract agar medium. The entire mycelium was separated from at least 3 plates per concentration and supplement after 28 days using a scalpel and stored and ground as described above. The mycelium of the plate supplemented with 50 mM L-cysteine was harvested after 33 days due to reduced growth.

DNA-based determination of the fungal species via internal transcribed spacer (ITS) sequencing

Genomic DNA was isolated from fresh mycelium according to the protocol described by Liu et al. (2000). The initial cell disruption step was performed using glass beads ($\varnothing$ 0.25–0.50 mm, Carl Roth, Karlsruhe, Germany) added to the lysis-buffer-mycelium-mixture and a subsequent shaking (5 min, 30 Hz, room temperature) with the MM 400 mixer mill (RETSCH, Haan, Germany) before incubating the sample for 10 min.

The concentration of the obtained DNA was determined using the NanoDrop Lite Plus UV/Vis spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA), set to a concentration between 10 and 100 ng/ μ L with pure water and used for polymerase chain reaction (PCR) to amplify the 5.8S rRNA gene sequence and the adjacent Internal Transcribed Spacer (ITS) regions. For more detailed information on the PCR performed, see supporting information Tables S1 and S2. The amplicon was purified with the NucleoSpin Gel and PCR Clean-Up Mini Kit (Macherey–Nagel, Düren, Germany) according to the manufacturer's protocol except for the elution, which was done with 50 μ L pure water. The purified sample was sent for Sanger sequencing (Eurofins Genomics, Ebersberg, Germany) with the ITS4- (5'-TCC TCC GCT TAT TGA TAT GC-3') and ITS5-primers (5'-GGA AGT AAA AGT CGT AAC AAG-3') according to the contractor's sample specifications. The sequencing data were assessed using Geneious 11.1.5 (supporting information Figure S1) and compared to the Nucleotide Collection (nr/nt) database using BLAST (max. *E*-value, 10^{-10} ; <https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE=Nucleotides>, last accessed July 2026).

Nutritional analysis

The percentage of moisture was assessed utilizing a DAB 100–3 moisture analyzer (Kern and Sohn GmbH, Balingen-Frommern, Germany). The concentrations of crude fat (Weibull-Stoldt acid digestion and extraction using petroleum ether), crude ash (incineration at 550 °C), crude protein (calculated via the Kjeldahl method with a conversion factor of 6.25), and carbohydrates (by difference formation), along with the amino acid profile, were evaluated following protocols outlined by Ahlborn et al. (2019). For the amino acid analysis, sample weights and volumes were reduced to one-tenth of those specified in Commission Directive 98/64/EC of the European Union. β -Glucan levels were measured using the β -glucan Assay Kit (Yeast and Mushroom) from Megazyme (Wicklow, Ireland) and total dietary fiber content was quantitated using the Total Dietary Fiber Assay Kit provided by Merck (Darmstadt, Germany). The fatty acid composition was analyzed through direct transmethylation with methanolic potassium

hydroxide, as detailed by Hammer et al. (2021). Fatty acid analysis was performed using gas chromatography mass spectrometry (GC–MS; Agilent 7890 A GC system coupled with a 5975 C mass spectrometry detector; Agilent Technologies, Waldbronn, Germany) and an Agilent J&W VF-WAXms capillary column (30 m $\times$ 0.25 mm inner diameter, 0.25 μ m film thickness). The instrumental parameters were set as follows: helium was used as the carrier gas at a constant flow rate of 1.2 mL/min; the inlet temperature was maintained at 250 °C with an injection volume of 1 μ L and a split ratio of 20:1; the septum purge flow rate was set to 3 mL/min; the scan range was *m/z* 33–300; electron ionization energy was 70 eV; the source temperature was 230 °C; the quadrupole temperature was 150 °C; and the transfer line temperature was 250 °C. The oven temperature program consisted of an initial hold at 40 °C for 3 min, followed by a ramp at 5 °C/min to 240 °C, which was then held for 12 min. A fatty acid methyl ester (FAME) standard (C4–C22), sourced from CS-Chromatographie Service GmbH (Langerwehe, Germany), was used to identify the peaks. By evaluating the respective integration areas, the relative proportions of FAME in the sample were determined.

To assess macronutrients (Ca, K, Mg, Na, P) and trace elements (B, Cu, Fe, Mn, Zn), 0.3 g of each sample was digested using 6 mL of 65% HNO₃ and 1.5 mL of 30% H₂O₂ in a microwave system (Mars 6 One Touch Express, CEM Corporation, Matthews, USA). The following digestion program was applied: a temperature of 180 °C, pressure of 800 psi, and power ranging from 900 to 1800 W; the temperature was ramped to 140 °C over 20 min, then to 180 °C over 15 min, and held at 180 °C for 15 min, followed by a 20-min cooling period. After filtration and adjustment to a volume of 50 mL with ultra-pure water, the concentrations of the analytes in the digested samples were determined using Inductively Coupled Plasma Optical Emission Spectroscopy (ICP-OES, Avio 220 Max; Perkin Elmer, Shelton, USA).

The energy content was computed using the following factors as described by Teke et al. (2021): 4 kcal/g for protein, 9 kcal/g for fat, and 4 kcal/g for available carbohydrates. Available carbohydrates were calculated as 100 minus the sum of proteins, fat, ash, and total dietary fiber content. All results are expressed on a dry matter (DM) basis, unless otherwise specified.

Aroma analysis and quantitation

To carry out HS-SPME analysis, dried surface mycelium (30 mg) and 3 mL of fully demineralized water were weighed into 20 mL headspace vials and sealed with silicon/PTFE screw caps. A 10-min incubation at 40 °C and 250 rpm was performed using an MPS multipurpose sampler (Gerstel, Mülheim an der Ruhr, Germany). Extraction was then performed for 30 min under the same conditions, using a DVB/C-WR/PDMS (80 μ m) SPME fiber (Supelco,

Steinheim, Germany). For the analysis of the fruiting bodies, a distinction was made between the analysis of fresh and dried fruiting bodies on the one hand, and between the addition of water and no addition of water in the vial on the other. In each case, 100 mg of mushroom was weighed in, with the fresh fruiting bodies being cut into 1 mm pieces immediately prior to analysis. 0.2 µg of internal standard diallyl disulfide (DADS; stock solution in ethanol: 2000 µg/mL) was added to each vial prior to incubation. Separation was achieved using the same GC system as described above, with the liner adapted to SPME. For the analysis, the surface mycelia were analyzed in splitless mode, while the fruiting bodies were analyzed using a split ratio of 2:1.

Compound identification was performed as published by Ahlborn et al. (2026) by measurement on both a polar and a nonpolar column using standards, blanks, retention indices, MS spectra, and olfactometry.

Quantitation was performed using the internal standard diallyl disulfide. Of the identified aroma compounds, the amounts could be quantitated by calculating the response factor (Rf) for 2,3,5-trithiahexane (2,3,5-TTH) and dimethyl trithiocarbonate (DMTTC); for all other aroma compounds, the amounts were semi-quantitated. 2,3,5-TTH was determined in a range from 0.01 to 0.5 µg absolute against an absolute amount of 0.2 µg internal standard in each vial (R^2 , 0.9980; Rf, 2.99 ± 0.24). For DMTTC, the range was 0.01 to 0.1 µg (R^2 , 0.9460; Rf, 1.54 ± 0.25). As with the samples, the final volume in the headspace vials was 3 ml, and they were filled to volume with water. Stock solutions were prepared in ethanol (2,3,5-TTH, 1000 µg/mL; DMTTC, 10 µg/mL).

Statistics

All analyses of nutrient composition were performed at least twice. In cases of double determination, half ranges were given as deviations; in cases of triple determination, the standard deviation was given. All analyses for aroma quantitation were performed thrice. To determine significant differences in aroma quantitation, one-factor analyses of variance (with alpha 0.05) were performed.

Results

Fungal species determination using ITS

Sequencing of the complete 5.8S rRNA gene with the adjacent partial ITS1 and ITS2 regions (supporting information Figure S1) confirmed the fungus as *Mycetinis scorodonius*. Despite repeated sequencing, the nucleobase at position 615 could not be clearly identified as T or C, as both nucleobases were present in equal proportions. This leads to the conclusion that the mycelium is dikaryotic. This was confirmed under the microscope by

Table 1 Nutritional composition of *Mycetinis scorodonius* fruiting bodies (results on dry weight basis except for moisture content). *FB* fruiting bodies

Parameter	Amount [g/100 g]
Moisture content	
Fresh FB	78.80 ± 0.93
Freeze-dried FB	9.03 ± 0.04
Crude protein	31.02 ± 0.48
Crude fat	< 0.5
Crude ash	5.28 ± 0.01
Carbohydrates	63.44 ± 0.70
Dietary fiber	38.65 ± 0.07
Insoluble fiber	38.65 ± 0.07
Soluble fiber	nd
Total glucans	18.06 ± 0.25
α -glucans	3.94 ± 0.07
β -glucans	14.92 ± 0.31
Energy [kcal/100 g DM]	225 ± 5

nd not detectable

Table 2 Mineral composition of *Mycetinis scorodonius* fruiting bodies analyzed by ICP-OES. *DM* dry matter

Minerals	Amount [mg/100 g DM]	Minerals	Amount [mg/100 g DM]
Ca	nd	B	nd
K	1103 ± 4.7	Cu	1.2 ± 0.0
Mg	115 ± 12.1	Fe	3.5 ± 0.8
Na	nd	Mn	2.3 ± 0.0
P	760 ± 1.8	Zn	10.0 ± 0.2

nd not detectable

the presence of clamp connections (see supporting information Figure S2), which are only found in dikaryons (Furtado 1966).

Nutritional composition of *M. scorodonius* fruiting bodies

The summary of the proximate analysis of dried *M. scorodonius* fruiting bodies (Table 1) shows a high protein content and a high dietary fiber content, with both contents each accounting for more than 30% on a dry weight basis. The mineral content, amino acid profile, and fatty acid profile are shown in Tables 2, 3, and 4, revealing a high potassium content of 1103 mg/100 g dry matter (DM), glutamic acid/glutamine as the main amino acid, and linoleic acid as the main fatty acid. The total amount of all minerals analyzed was 1.99% DM. The total amount of all amino acids was 25.97% DM, with an average nitrogen content in the protein of 17.49%.

Table 3 Amino acid composition of *Mycetinis scorodonius* fruiting bodies. DM dry matter

Amino acids	Amount [g/100 g DM]
Alanine	1.48 ± 0.03
Arginine	1.29 ± 0.03
Aspartic acid/asparagine	2.24 ± 0.05
Cysteine	0.23 ± 0.01
Glutamic acid/glutamine	5.42 ± 0.12
Glycine	1.06 ± 0.02
Histidine	3.74 ± 0.09
Isoleucine	0.98 ± 0.02
Leucine	1.54 ± 0.03
Lysine	1.47 ± 0.03
Methionine	0.34 ± 0.02
Phenylalanine	0.99 ± 0.02
Proline	0.90 ± 0.02
Serine	1.21 ± 0.02
Threonine	1.19 ± 0.03
Tryptophan	0.09 ± 0.02
Tyrosine	0.66 ± 0.01
Valine	1.14 ± 0.02

Table 4 Fatty acid profile of *Mycetinis scorodonius* fruiting bodies analyzed by GC–MS

Fatty acids	Relative proportion [%]
C14:0	0.87
C16:0	25.05
C18:0	9.99
C18:1	7.47
C18:2	56.63

Aroma composition of fruiting bodies

In previous studies (Ahlborn et al. 2026), nine sulfur-containing aromas were analyzed in freeze-dried fruiting bodies using HS-SPME: methanethiol, carbon disulfide, dimethyl disulfide, methyl dithioformate (MDTF), 2,4-dithiapentane, dimethyl trisulfide, 2,3,5-trithiahexane (2,3,5-TTH), dimethyl trithiocarbonate (DMTTC), and 2,4,5,7-tetrathiooctane (2,4,5,7-TTO). Since MDTF and 2,3,5-TTH contributed most to the aroma impression of the mushroom (Ahlborn et al. 2026), these were quantitated in the current study using HS-SPME. In addition, DMTTC and 2,4,5,7-TTO were also quantitated, as they were also present in the mycelia and 2,4,5,7-TTO is the reduced precursor molecule marasmin (2,4,5,7-TTO 4-oxide). The other sulfur-containing compounds were of minor importance, as the peak areas were very small and did not contribute significantly to the overall aroma impression (Ahlborn et al. 2026).

When examining the aroma profile of the fruiting bodies of *M. scorodonius*, significant differences were found in the amount, both when comparing fresh and dried fruiting bodies and when comparing water-activated and non-activated fruiting bodies, particularly in the case of 2,3,5-TTH. It was generally found that fresh fruiting bodies contained significantly more of the analyzed aroma compounds than dried fruiting bodies (Fig. 3). In the case of 2,3,5-TTH, only 28 mg/kg DM could be analyzed in the dried and activated fruiting bodies, compared to the fresh ones with 645 mg/kg DM. It is particularly notable that only a small amount of 2,3,5-TTH was found as the sole sulfur-containing compound in the lyophilized and non-activated fruiting bodies (1.5 mg/kg DM). This leads to the conclusion that *M. scorodonius* mushrooms can only develop their full flavor in a moist environment through enzyme activation.

Growth and aroma quantitation of surface mycelium of different ages

Young mycelium was white in color and had a fluffy, voluminous structure. As it aged, the structure changed to become more fibrous, hard, and less voluminous. The color changed from white to brown to dark brown, starting from the inoculum (Fig. 2B). This brown discoloration of *Mycetinis* mycelium has already been observed decades ago (Badcock 1939). However, the aroma of the mycelium and its dependence on age had not yet been investigated.

The mycelia of different ages, as described in the methods section, were examined for the four aroma compounds that had already been quantitated in the fruiting bodies. No other sulfur compounds were found in the mycelium either. The results showed that, with increasing age, the 2,3,5-TTH content in particular decreased significantly (Fig. 4). A decreasing trend with increasing age was also observed for DMTTC and 2,4,5,7-TTO. Only the levels of MDTF did not decrease continuously, and the differences were not significant.

Growth and aroma quantitation of mycelium grown on supplemented media

In order to compare the growth of the supplemented plates with each other, control plates with unsupplemented malt extract agar were prepared and treated in the same way as all other plates. These control plates showed radial growth, which covered over 80% of the plate after 13 days. Here, the first slight brown discoloration originating from the inoculum was already visible (Fig. 5). After 20 days of growth, the plates were more than 90% covered with mycelium and most of the mycelium had turned brown.

Compared to the control plates, the plates supplemented with 5 mM methionine (Fig. 5) showed a reduced brown

Fig. 3 Quantitated amounts of the aroma compounds 2,3,5-trithiahexane (2,3,5-TTH), methyl dithioformate (MDTF), dimethyl trithiocarbonate (DMTTC), and 2,4,5,7-tetrathiaoctane (2,4,5,7-TTO) in fresh and dried *M. scorodoni* fruiting bodies, as well as in fruiting bodies activated and not activated with water. Analyzed using HS-SPME GC-MS via an internal standard; MDTF and 2,4,5,7-TTO were only semi-quantitated. *FB* fruiting bodies

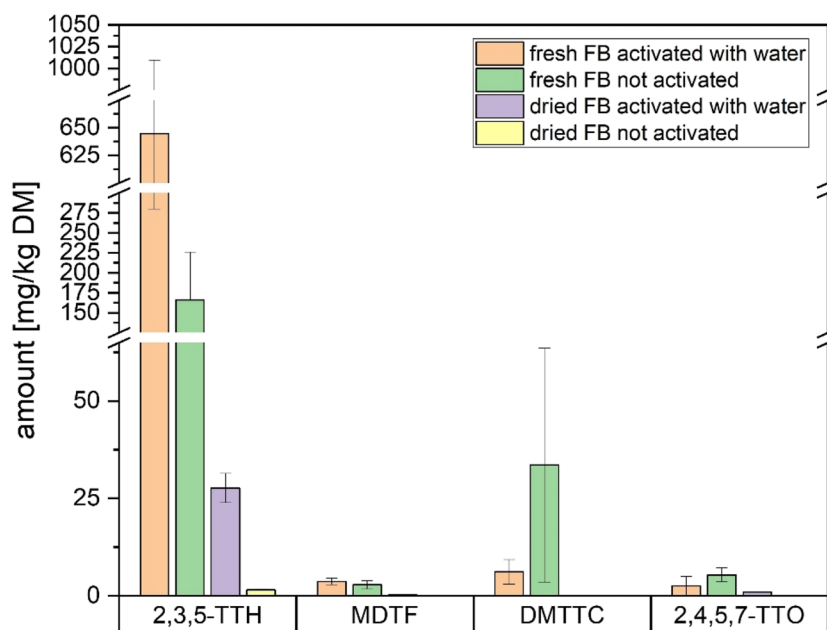
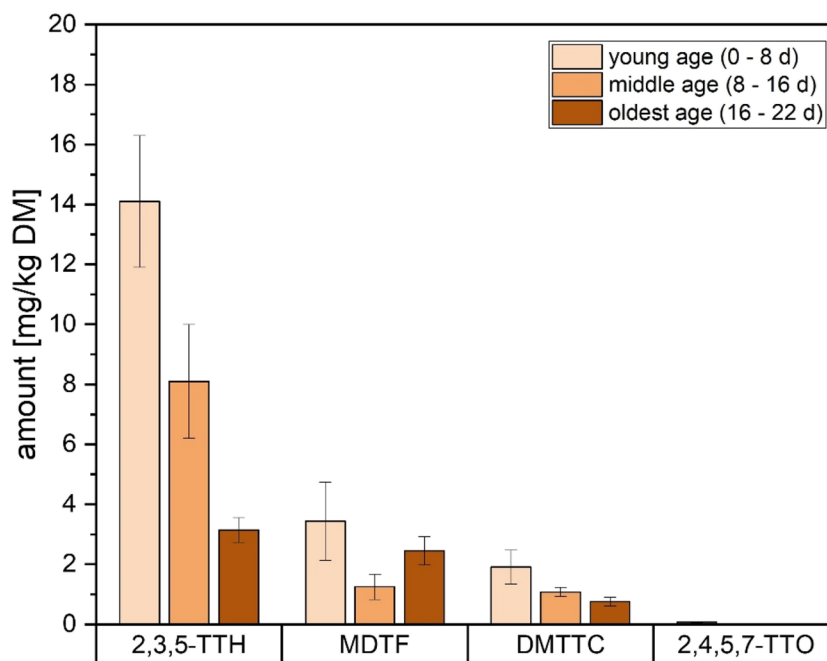


Fig. 4 Quantitation of aroma compounds found in *M. scorodoni* mycelium depending on the mycelium age. 2,3,5-TTH 2,3,5-trithiahexane, MDTF methyl dithioformate, DMTTC dimethyl trithiocarbonate, and 2,4,5,7-TTO 2,4,5,7-tetrathiaoctane



discoloration. After 20 days, the plates were completely covered, similar to the control plate.

Sulfate supplementation with increasing concentrations (Fig. 6) did not slow down growth. But at higher sulfate concentrations, the mycelium turned yellowish at some points before turning brown.

In contrast, supplementation with L-cysteine (Fig. 7) resulted in a strong inhibition of growth with increasing concentrations. This was accompanied by a reduction in the brown discoloration of the mycelium. In the case of

supplementation with 50 mM L-cysteine, growth only began after more than 20 days, and after 33 days amounted to only approximately 70% coverage of the plate.

To investigate the influence of the supplements on aroma formation, the levels of the four aroma compounds examined were analyzed. Supplementation with methionine (Fig. 8) led to a significant decrease in 2,3,5-TTH, MDTF, and 2,4,5,7-TTO levels, whereas DMTTC levels increased significantly. The addition of 5 mM sulfate (Fig. 8) was the only sulfate concentration at which the content of 2,3,5-TTH,

Fig. 5 Growth of *M. scorodonium* on malt extract agar plates over a period of 20 days, supplemented with 0 mM (control) and 5 mM L-methionine. Growth was only slightly slowed down in the beginning by the addition of L-methionine, but the brown discoloration was greatly reduced

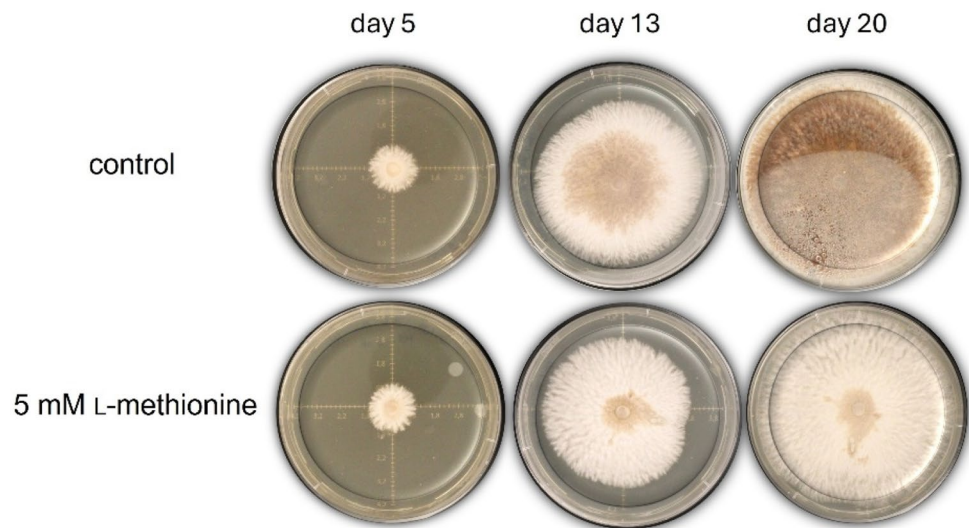
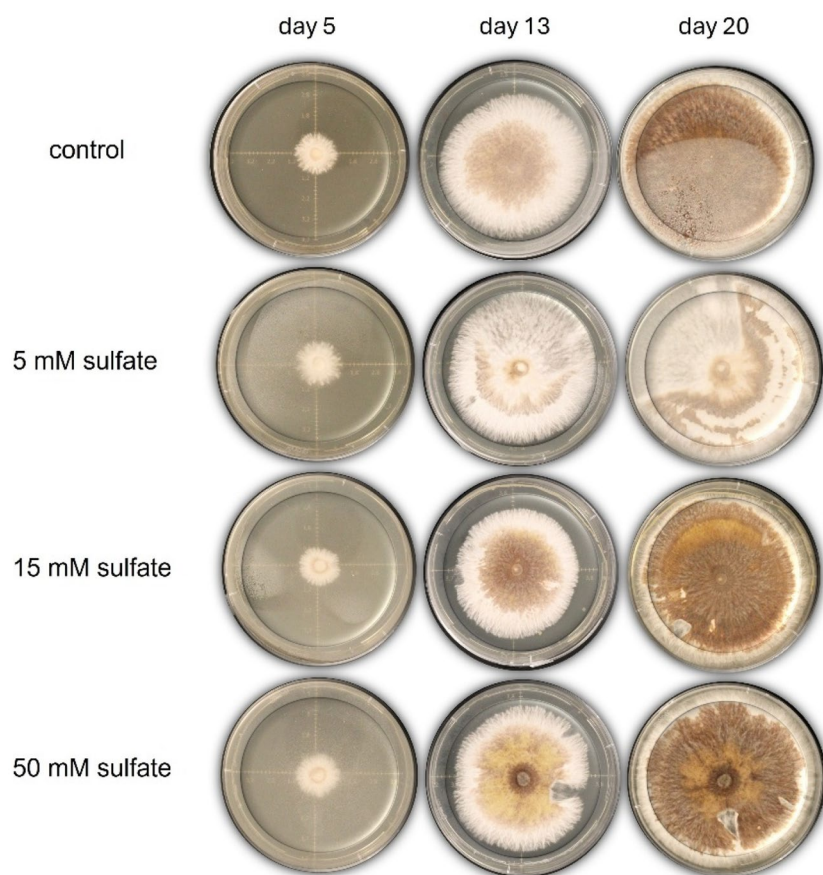


Fig. 6 Growth of *M. scorodonium* on malt extract agar plates over a period of 20 days, supplemented with different concentrations of sulfate: 0 (control), 5, 15, and 50 mM. Growth was not slowed down by increasing sulfate concentrations. At higher sulfate concentrations, the mycelium turned yellowish at some points before turning brown

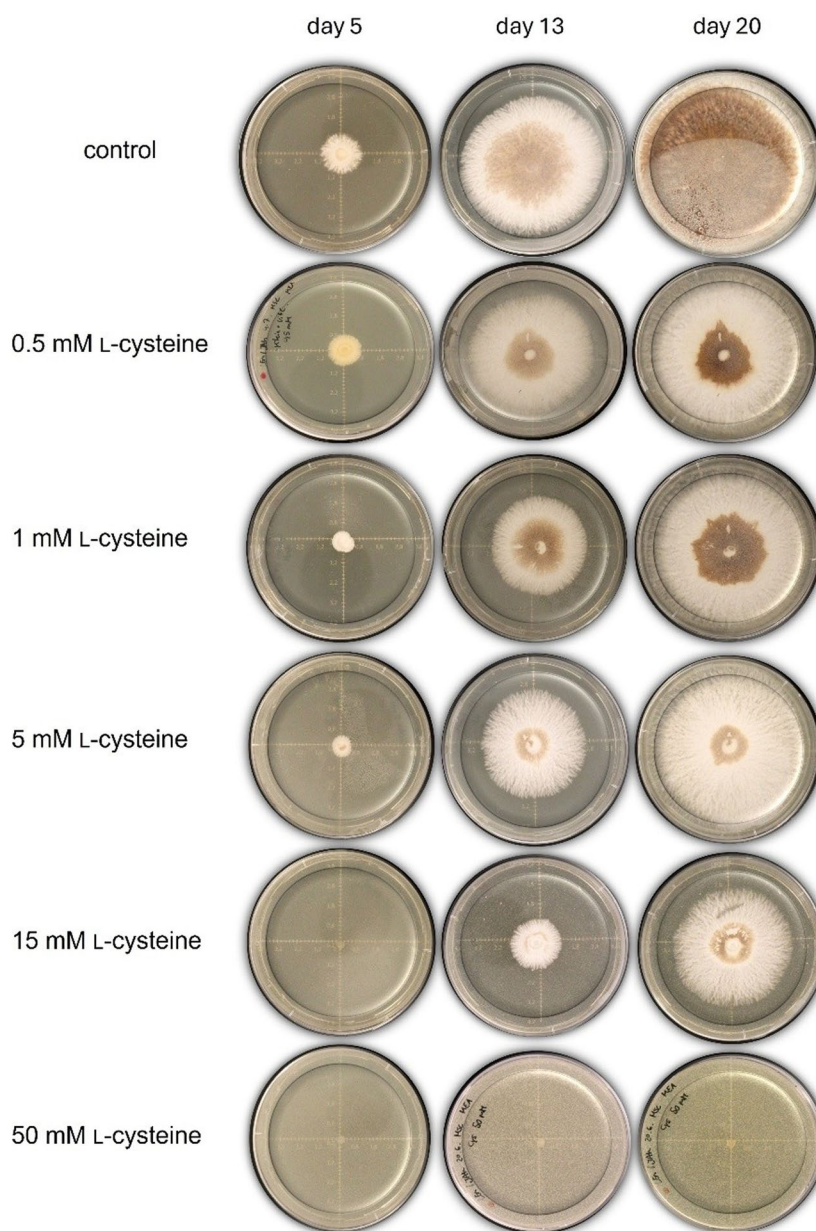


MDTF, and DMTTC increased significantly, although the increase did not even double the amount of 2,3,5-TTH. Higher concentrations of sulfate led to decreases in all four aroma compounds.

Supplementation with L-cysteine (Fig. 9), on the other hand, had a much more pronounced effect on 2,3,5-TTH.

Here, the amount of 6 mg/kg DM in the control could be increased to 66 mg/kg DM with a 5 mM L-cysteine supplement. All supplemented L-cysteine concentrations led to a significant increase in the 2,3,5-TTH content compared to the control. The levels of 2,3,5-TTH analyzed after supplementation with 2, 5, and 15 mM L-cysteine are all insignificant

Fig. 7 Growth of *M. scorodoni* on malt extract agar plates over a period of 20 days, supplemented with different concentrations of L-cysteine: 0 (control), 0.5, 1, 5, 15, and 50 mM. Growth rates decreased with increasing L-cysteine concentration and the brown discoloration was greatly reduced



when compared to each other. However, compared to 1 mM L-cysteine supplementation, they are significantly increased. Of the possible precursors examined here, L-cysteine therefore had the greatest enhancing effect on aroma content.

Discussion

Nutritional composition

To the best of our knowledge, no data on the nutrient composition of *M. scorodoni* is available in the literature. Therefore, the analyzed composition was compared with literature values for fungi from the families in which *M.*

scorodoni has been classified over recent decades. *M. scorodoni* is currently classified as belonging to the family *Omphalotaceae*. *Lentinula edodes* was selected as the comparative fungus here. Before *M. scorodoni* belonged to the *Omphalotaceae*, it was classified as part of the *Marasmiaceae* and, much earlier, as part of the *Tricholomataceae*. Data from *Marasmius oreades* and *Lepista nuda* (synonym: *Collybia nuda*) were used for comparison with these earlier families.

Comparison of the composition of the main nutrients showed good agreement with the current family *Omphalotaceae* (*Lentinula edodes*), particularly in terms of ash, fat, and carbohydrate content (Table 5). Our fiber data shows a significantly higher percentage compared to the values cited

Fig. 8 Quantitation of aroma compounds in the mycelium of *M. scorodoni* grown on agar plates supplemented with L-methionine or sulfate. 2,3,5-TTH 2,3,5-trithiahexane, MDTF methyl dithioformate, DMTC dimethyl trithiocarbonate, and 2,4,5,7-TTO 2,4,5,7-tetrathiaoctane. *: values differ significantly from the control

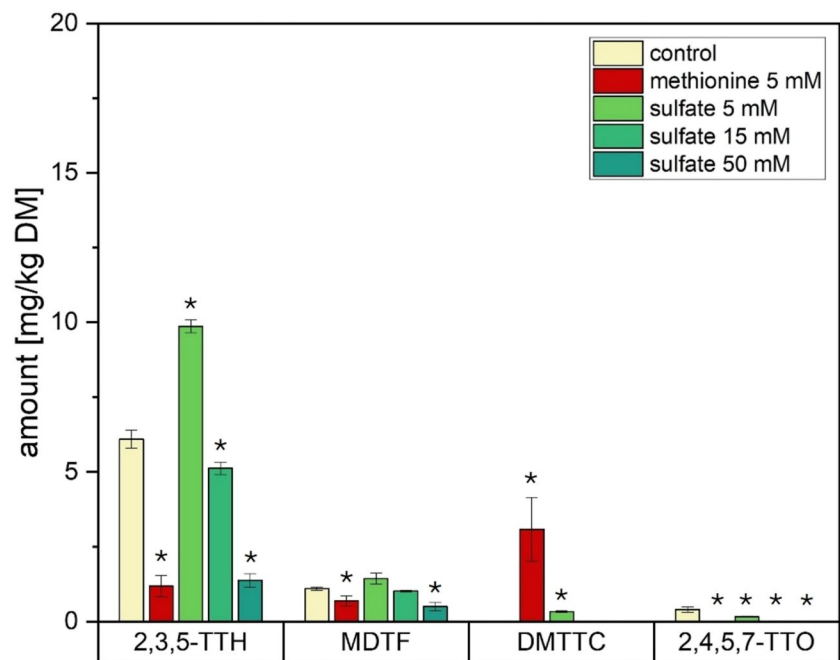
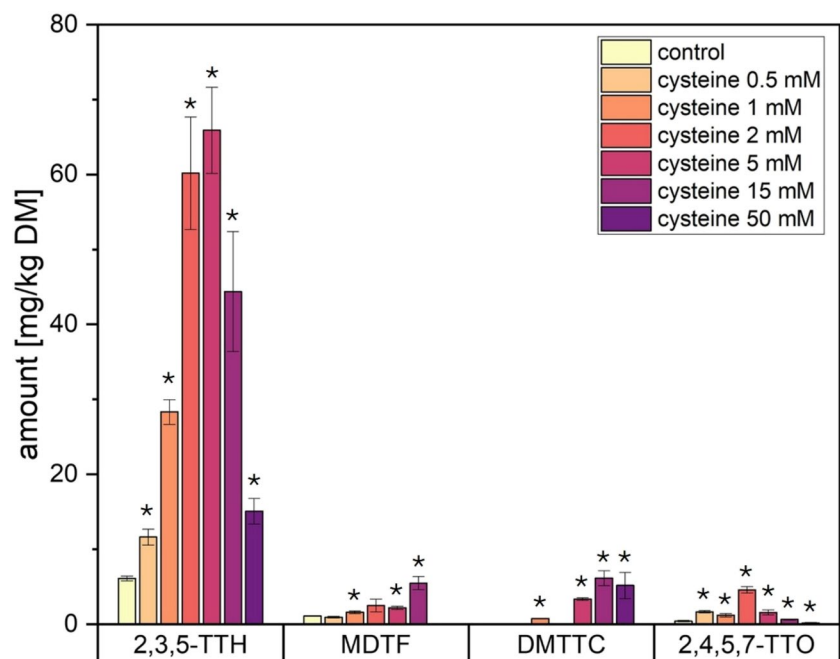


Fig. 9 Quantitation of aroma compounds in the mycelium of *M. scorodoni* grown on agar plates supplemented with L-cysteine. 2,3,5-TTH 2,3,5-trithiahexane, MDTF methyl dithioformate, DMTC dimethyl trithiocarbonate, and 2,4,5,7-TTO 2,4,5,7-tetrathiaoctane. *: values differ significantly from the control



in the literature. However, since the results of fiber analysis are highly dependent on the method used, a comparison is difficult, especially when the methods are not clearly specified. We used the Total Dietary Fiber Assay Kit from Merck, which is based on Lee et al. (1992). It is an enzymatic-gravimetric method that can also distinguish between soluble and insoluble fiber. The method involves enzymatic degradation

with heat-stable α -amylase, protease, and amyloglucosidase in MES-Tris buffer instead of phosphate buffer, followed by ethanol precipitation and correction for protein and ash residues. The authors of the data cited here for the families *Omphalotaceae* and *Marasmiaceae* wrote that they also used an enzymatic gravimetric method that utilized the three enzymes, but they also referred to an AOAC method from

Table 5 Comparison of the analyzed proximate composition for *M. scorodonius* with literature values from the three families into which *M. scorodonius* has been classified in recent years. All values are based on dry matter

Fungal species	<i>Mycetinis scorodonius</i>	<i>Lentinula edodes</i>	<i>Marasmius oreades</i>	<i>Lepista nuda</i>
Reference	This study	(Dimopoulou et al. 2022)	(Dimopoulou et al. 2022)	(Barros et al. 2008)*
Family	<i>Omphalotaceae</i>	<i>Omphalotaceae</i>	<i>Marasmiaceae</i>	<i>Tricholomataceae</i>
Crude protein [g/100 g]	31.0	20.7	38.5	59.4
Crude fat [g/100 g]	< 0.5	1.3	3.8	1.8
Crude ash [g/100 g]	5.3	6.1	1.3	18.5
Carbohydrates [g/100 g]	63.4	59.9	36.1	24.9
Dietary fiber [g/100 g]	38.7	3.8	11.6	na
Energy [kcal/100 g]	225	340	355	335

*Values were converted to a dry weight basis; na not available

Table 6 Comparison of the analyzed fatty acid composition of *M. scorodonius* with literature values from the three families into which *M. scorodonius* has been classified in recent years. Values given in relative proportions [%]

Fungal species	<i>Mycetinis scorodonius</i>	<i>Lentinula edodes</i>	<i>Marasmius oreades</i>	<i>Lepista nuda</i>
Reference	This study	(Mizuno 1995)	(Yilmaz 2025)	(Barros et al. 2008)
Family	<i>Omphalotaceae</i>	<i>Omphalotaceae</i>	<i>Marasmiaceae</i>	<i>Tricholomataceae</i>
C14:0	0.9	0.3	0.5	0.3
C16:0	25.1	19.1	33.0	11.8
C18:0	10.0	0.7	18.9	2.4
C18:1	7.5	4.8	18.1	29.5
C18:2	56.6	71.0	29.0	51.5

1970 where sulfuric acid and sodium hydroxide are used (Dimopoulou et al. 2022). It therefore remains unclear which parameters were used exactly, making a comparison difficult.

Compared to the family *Marasmiaceae*, the protein content is similar, but the carbohydrate content differs by 27 g/100 g. The differences from the original first family of *Tricholomataceae* (*Lepista nuda*) are very large. In particular, the protein, ash, and carbohydrate content differ greatly (Table 5).

When comparing the fatty acid profiles, the distribution also corresponds well with that of *Lentinula edodes* and less with *Marasmius orades* and *Lepista nuda* (Table 6). The amino acid composition is also highly similar to that of *Lentinula edodes*; only histidine and tyrosine differ (Table 7). In both mushrooms, glutamic acid is the primary amino acid. When comparing mineral compositions, the greatest similarity is found with the family *Omphalotaceae*, particularly in terms of the high potassium content (Table 8). Potassium is the major mineral element in many mushrooms and often accounts for between 2 and 4% of the dry weight in wild mushrooms (Kalač 2009). As for trace elements, all four elements analyzed fall within the normal range for mushrooms (Kalač 2009).

The current classification within the family *Omphalotaceae* shows the least deviation overall, so that classification within the *Omphalotaceae* also appears reasonable from an analytical point of view.

Aroma content of fruiting bodies

The claim that the fruiting bodies can be dried without significant loss of aroma (Laux 2019) cannot be confirmed by the data analyzed here. Although the dried fruiting bodies still had a very intense flavor after rehydration, the fresh fruiting bodies far surpassed them in this regard, as clearly shown in Fig. 3. Different drying methods and drying times always affect the flavors, as is already well known in the literature (Pei et al. 2016; Politowicz et al. 2018; Zhang et al. 2021). Freeze-drying is considered among the best methods for drying fruiting bodies, as this method has been shown to preserve the highest levels of aroma and key aroma compounds as well as bioactive compounds due to low temperatures (Politowicz et al. 2018).

Sulfur compounds are typically formed through an enzymatic process, and therefore the aroma profile can vary at any stage of the drying process (Politowicz et al. 2018; Zhang et al. 2021). In shiitake mushrooms, these include, for example, 1,2,4-trithiolane, 1,2,4,5-tetrathiane, 1,2,3,5,6-pentathiepane and hexathiepane (Politowicz et al. 2018). They found that when shiitake mushrooms were dried using freeze-drying, the content of characteristic aromas decreased by approximately 50% compared to fresh fruiting bodies, which represented the smallest loss. Other methods, such as convective pre-drying in combination with vacuum-microwave drying, suffered a 90% loss, by contrast. And that is

Table 7 Comparison of the analyzed amino acid composition of *M. scorodonius* with literature values. Values given in g/100 g dry matter

Fungal species	<i>Mycetinis scorodonius</i>	<i>Lentinula edodes</i>	<i>Marasmius oreades</i>	<i>Lepista nuda</i>
Reference	This study	(Mattila et al. 2002)*	(Leichter and Bandoni 1980)**	(Ayaz et al. 2011)
Family	<i>Omphalotaceae</i>	<i>Omphalotaceae</i>	<i>Marasmiaceae</i>	<i>Tricholomataceae</i>
Alanine	1.48	1.24	2.49	1.29
Arginine	1.29	1.51	2.12	0.91
Aspartic acid/asparagine	2.24	2.26	3.95	1.76
Cysteine	0.23	0.29	0.31	0.34
Glutamic acid/glutamine	5.42	4.23	4.46	2.85
Glycine	1.06	1.08	2.07	0.78
Histidine	3.74	0.67	0.89	0.68
Isoleucine	0.98	0.94	1.96	1.06
Leucine	1.54	1.58	3.17	1.38
Lysine	1.47	1.45	2.64	1.07
Methionine	0.34	0.35	0.59	0.28
Phenylalanine	0.99	1.08	1.70	0.81
Proline	0.90	1.07	1.49	0.73
Serine	1.21	1.23	2.54	0.94
Threonine	1.19	1.17	2.10	1.01
Tryptophan	0.09	na	0.59	0.28
Tyrosine	0.66	3.15	1.64	0.31
Valine	1.14	1.48	1.73	0.80

*Values were converted to a dry weight basis; **Original values were given in g/100 g protein, which were converted here to g/100 g dry matter using moisture and protein content; na not available

Table 8 Comparison of the analyzed mineral composition of *M. scorodonius* with literature values. Values given in mg/100 g dry matter

Fungal species	<i>Mycetinis scorodonius</i>	<i>Lentinula edodes</i>	<i>Marasmius oreades</i>	<i>Lepista nuda</i>
Reference	This study	(Siwulski et al. 2021)	(Genççelep et al. 2009)	(Genççelep et al. 2009)
Family	<i>Omphalotaceae</i>	<i>Omphalotaceae</i>	<i>Marasmiaceae</i>	<i>Tricholomataceae</i>
Ca	nd	27	713	880
K	1103	1640	170	212
Mg	115	102	454	341
Na	nd	13	485	62
P	760	337	249	278
Cu	1.2	0.4	0.9	2.7
Fe	3.5	2.7	30.5	84.2
Mn	2.3	1.6	2.7	5.9
Zn	10.0	5.8	6.1	7.6

nd not detectable

also the range we observed for *M. scorodonius* fruiting bodies after freeze-drying: For 2,3,5-TTH, for example, the drying process resulted in a loss of 96%. In other words, even though a gentle drying method was used, there was a significant loss of aroma compounds. The reasons for this could include the inactivation of enzymes by the drying process, as well as the instability of sulfur compounds. Schmidberger and Schieberle (2020) found that frying shiitake mushrooms produced new flavor compounds that were not present in raw

shiitake mushrooms. Consequently, not only does the drying method have a significant impact on the flavor, but so does the preparation method.

The sometimes large standard deviations observed for certain aroma compounds in the fresh fruiting bodies of *M. scorodonius* (Fig. 3) can be explained by the fact that different fruiting bodies had to be used for each individual measurement in the triplicate analysis, since the fruiting bodies were fresh. Although fruiting bodies of similar size

and ripeness were used for the analysis, even these can vary significantly. Triple determination using the same fresh fruiting body was not possible, as harvesting and damaging the mushroom would lead to a loss of volatiles and dehydration. In addition, the stipe base, stipe, and cap may contain varying levels of aroma compounds, as has been described for some other mushrooms (Cho et al. 2008; Orban et al. 2020). In contrast, the analyses of the dried and homogenized *M. scorodוניus* fruiting bodies showed only small standard deviations.

Gmelin's observation that a stronger garlic aroma is released upon contact with water (Gmelin et al. 1976) is clearly supported by our data. This holds true for both fresh fruiting bodies and, in particular, for dried fruiting bodies, which released hardly any sulfur compounds without the addition of water (Fig. 3). This is likely due to the enzymes involved, which require an aqueous environment. Enzymes already known to be involved in the formation of marasmin from γ -glutamylmarasmin include γ -glutamyl transpeptidase and a C-S lyase (Gmelin et al. 1976). No other enzymes involved in the formation of γ -glutamylmarasmin are currently known. However, we are investigating this topic at the moment. In the case of garlic, the aroma compounds can only be released once the cells are broken down, since the substrate and enzyme are located in separate compartments (alliin in the cytoplasm and alliinase in the vacuoles) (Block 1992). This could also be the case with *Mycetinis*. However, the exact location has not yet been investigated.

Aroma content of the mycelium

Aroma analyses of mycelium of varying ages revealed that the aroma compounds diminish with age. This finding contradicts the results of Badcock (1939), who observed that white mycelium of a 25-day-old *M. alliaceus* culture is odorless, while brown mycelium has a strong garlic odor. In older cultures, however, the odor of white mycelium would also increase. This data was apparently determined based on olfactory impressions rather than detected using analytical instruments, which can be attributed to the age of the publication. At that time, the brown discoloration of the mycelium had already been observed, though its cause, or the conditions under which it occurred, remains unclear to this day.

L-Cysteine supplementation yielded very clear results, leading to an 11-fold increase in 2,3,5-TTH levels (Fig. 9), which is one of marasmin's breakdown products (Ahlborn et al. 2026). It therefore stands to reason that cysteine is directly involved in the formation of the flavor precursor γ -glutamylmarasmin. This would also be consistent with the pathways of alliin formation in garlic described in the literature. Yoshimoto and Saito (2019) investigated the formation of alliin and described the putative biosynthetic pathway starting with glutathione biosynthesis and valine catabolism. Glutathione biosynthesis requires L-cysteine, as

it is an intrinsic component (γ -glutamylcysteinylglycine). The biosynthesis of γ -glutamylmarasmin in the fungus could also involve glutathione, which would require further investigation.

Supplementation with 5 mM sulfate results in only a slight increase. Fungi can utilize sulfate as a source of sulfur by reducing it to hydrogen sulfide (sulfate assimilation pathway) and incorporating it into the *O*-acetylhomoserine pathway, which ultimately leads to the formation of homocysteine, which in turn can be converted to cysteine via the transsulfuration pathway (Toh-e et al. 2018). It is therefore reasonable to expect that the increase is less pronounced than with L-cysteine supplementation, since other substances, such as homoserine, must first be formed and must be present in sufficient quantities. Furthermore, this conversion takes more time than if cysteine were added directly and could be utilized immediately.

L-Methionine supplementation did not result in an increase in flavor compounds, suggesting that it does not play a direct role in the biosynthetic pathway. Nevertheless, methionine may be important for the formation of γ -glutamylmarasmin, since, for example, enzymes such as methyltransferases require *S*-adenosylmethionine (SAM) as a cofactor (Bennett et al. 2017). SAM, in turn, is synthesized from methionine. Methyltransferases transfer methyl groups that may be involved in the formation of the (methylthio)methyl side chain of γ -glutamylmarasmin.

It is striking that significantly more dimethyltrithiocarbonate was formed after methionine supplementation than without supplementation (Fig. 8). This might be because methionine can be demethylated to the thiol homocysteine, which can then react with carbon disulfide, which is also present in the mushroom (Ahlborn et al. 2026) to form trithiocarbonates (Souza et al. 2017). Dimethyltrithiocarbonate could then be formed through rearrangements.

The biosynthetic pathway for the formation of γ -glutamylmarasmin has not yet been studied. In the future, however, we will examine this in greater detail and attempt to elucidate the biosynthetic pathway of γ -glutamylmarasmin. The knowledge gained here serves as a starting point for a possible pathway.

Conclusion and outlook

In conclusion, based on its nutritional composition and comparisons with previous families, *M. scorodוניus* is best classified within the current family *Omphalotaceae*.

The data show that there are indeed significant differences in the levels of the analyzed aroma compounds between fresh and dried fruiting bodies. In the case of mycelium, not only does age affect the levels, but supplementation can also have a significant impact. In particular, the addition of L-cysteine resulted in an 11-fold increase in the

2,3,5-trithiahexane content. These findings can serve as a starting point for investigating the biosynthetic pathway of γ -glutamylmarasmin in *Mycetinis*, which has not yet been elucidated.

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Data availability The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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